

AI Resume Analyzer and Job Recommendation System

1-page project explanation for submission

Project Objective

I built a Python and Streamlit mini project that reads a PDF or DOCX resume, extracts the text, detects technical skills, compares the resume with job roles, and shows a simple learning roadmap for missing skills.

What I Have Done

Part	Work completed
Resume input	Added upload support for PDF and DOCX resumes and a file-size check.
Text processing	Cleaned extracted text using regular expressions so skill matching is more consistent.
Skill detection	Used a manually prepared skill dictionary and grouped detected skills by category.
Job matching	Implemented TF-IDF vector creation and cosine similarity to rank matching job roles.
Skill gap	Compared detected skills with required skills for the selected target role.
Roadmap	Generated a short week-wise roadmap for the missing skills.
Dashboard	Built a Streamlit interface with metrics, charts, recommendations, and report download.

Method Used

The resume and job role descriptions are converted into tokens. TF-IDF gives importance to useful words, and cosine similarity calculates how close the resume is to each job role. The app then shows the top matching roles and missing skills.

Files Prepared for Submission

The required project files are the Python modules, CSV datasets, sample resumes, README, project report, architecture diagram, tests, and requirements file. Generated Python cache files were removed because they are not needed for checking or running the project.

AI-Look Check and Responsible Use

The project title uses "AI" because the app performs resume analysis and recommendations. The implementation is simple and explainable: it uses CSV files, basic NLP cleaning, TF-IDF, and cosine similarity instead of hidden APIs. I also added clear comments in the code so a teacher can understand the logic. The app is for learning guidance only and should not be used for final hiring decisions.